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Name = Krish
Rollno = 2401010177
Course = Database Management System.

Assignment - 2

Section-A

Que.

(i) The Entity Relationship Model visually represents how entities relate in a database, offering a clear blueprint of the data's organizations structure.

(ii) In the Relational model, data is organized into tables with relationships b/w entities defined through foreign Key Constraints.

(iii) The object-oriented model treat data as a collection of objects, each with unique attribute & method, mirroring real-world entities & interactions.

(iv) Integrity Constraints ensure the reliability of data by enforcing rules that must be adhered. to, monitoring the accuracy of, information.

Section - B :-

Que2. In a healthcare system, discuss how data abstraction can simplify the interaction b/w healthcare professionals & patient records.

⇒ Data abstraction is the process of hiding complex database details and showing only relevant information to users. In a healthcare system, data abstraction helps doctors,

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nurses, and staff easily access patient information without dealing with complex database structures.

for ex. → Doctors can view patient medical history, diagnosis & prescriptions without seeing how the data is stored in the database

→ Nurses can update patient vitals & treatment details

→ Receptionists can easily access appointment & basic patient details.

Through abstraction level:-

(i) Physical level:- Shows how data is stored in the databases.

(ii) Logical level:- Shows the structure of the database like table & Relationships

(iii) View level:- Shows only the data required by specified users. This makes the system easier to use, secure of efficient for healthcare professionals.

Ques 3. Explore the potential benefits & challenges associated with implementing data abstraction in this context.

⇒ Benefits:-

(i) Security:- Sensitive patient information can the restricted to authorized users only.

(ii) Simplicity:- Healthcare staff can access required data without understanding database complexity.

(iii) Data Integrity:- Maintains accuracy & consistancy of patient data.

(iv) Efficiency:- Faster access to patient records improves medical decision making.

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Challenges:-

(i) Implementation Complexity: Designing multiple abstraction level can be difficult.

(ii) Performance overhead: Additional abstraction layers may slightly affect system performance.

(iii) Maintance Difficulty: Updating the system requires careful management of different views & structures.

Section - C (Simulation - based):

Ques 3. Design a database for a university, incorporating entities like students, course & professor, using the entity - relationship model, illustrate & explain the relationship b/w these entities include attribute for each entity to capture relevant details & professor information

⇒ Entities & Attributes:

(i) Student

Attributes

- Student_id (Primary Key)
- Name
- Age
- Email
- Department

(ii) Course:

- Course_ID (PK)
- Course_Name
- Credits
- Departments

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(3) Professor

- Professor - id (PK)
- Name.
- Email
- Department

Relationship:-

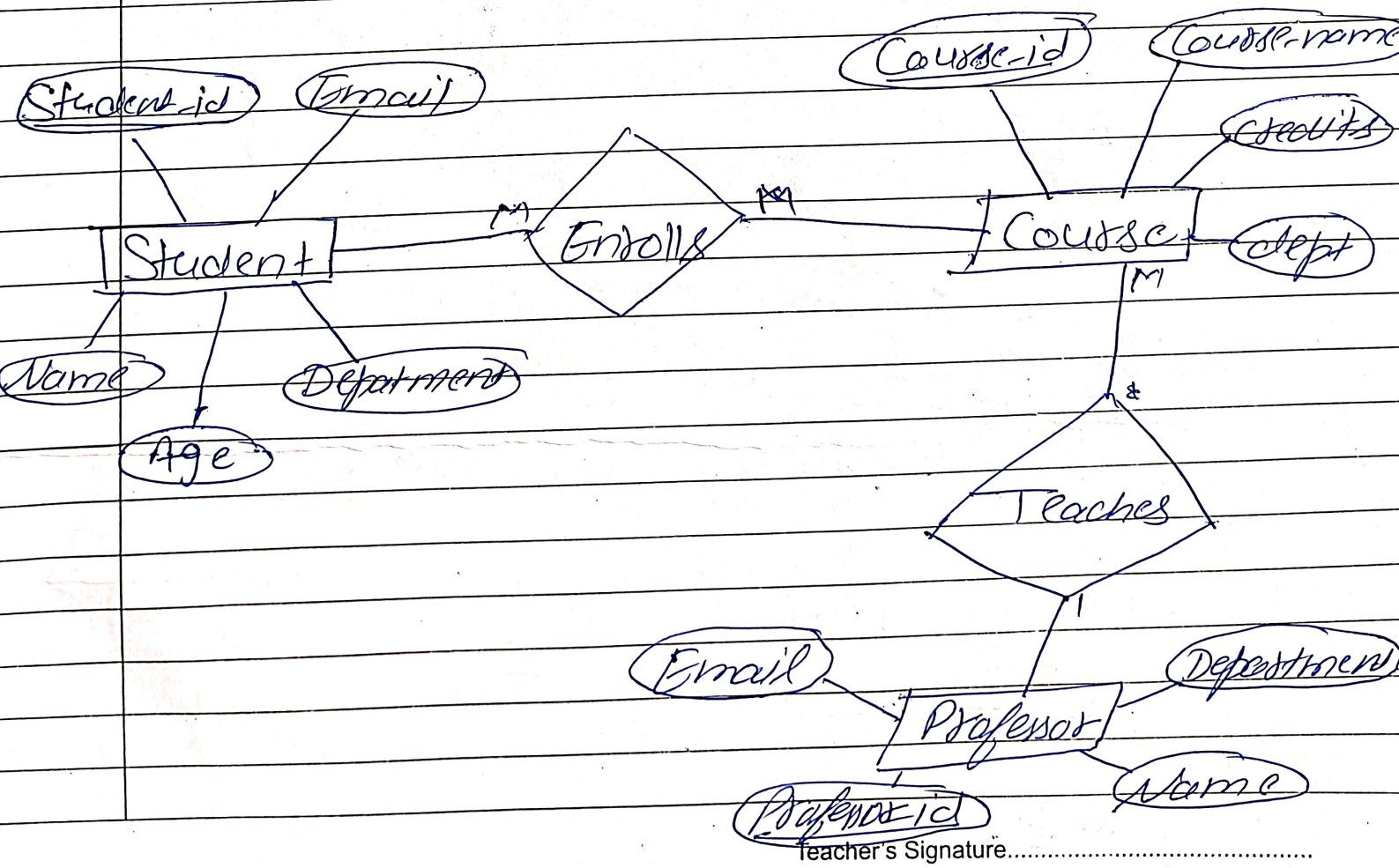
(i) Student - Enrolls - Course (M:N)

- One student can email in many course.
- one course can have many students.

(ii) Professor - Teaches - Course (1:M)

- one Professor can teach multiple student.
- Each Course is taught by one Professor.

Diagram:



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